



Patuakhali Science and Technology University

Faculty of Computer Science and Engineering

CIT 320 : Software Development Project-II

Project Proposal



Project Title : **MediPredict AI**

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MediPredict AI

AI-Based Multi-Disease Prediction and Health Monitoring Mobile Application

1. Introduction

The rapid advancement of artificial intelligence (AI) and mobile technology has significantly transformed the healthcare sector. Modern hospitals increasingly rely on digital systems to manage patient data, monitor health conditions, and assist medical professionals in making informed decisions.

However, many existing hospital management systems primarily focus on administrative tasks such as appointment scheduling and patient record management, while lacking intelligent health assessment features that can assist patients in early disease detection.

This project proposes the development of an **AI-Based Multi-Disease Prediction and Health Monitoring Mobile Application** using Flutter. The application will allow users to maintain their personal health profile, calculate Body Mass Index (BMI), and perform preliminary disease risk predictions using machine learning models. The system aims to help users monitor their health condition and receive early warnings regarding potential diseases such as heart disease, diabetes, liver disease, kidney disease, and others.

By integrating machine learning models with a mobile-based hospital management interface, the system will provide a convenient and accessible platform for health monitoring and disease risk assessment.

2. Objectives

The main objectives of this project are:

- To develop a **mobile-based healthcare monitoring system using Flutter**.
- To integrate **machine learning models for multi-disease prediction**.
- To provide **BMI and health-related calculations** for users.
- To create a **patient profile management system** containing important health data.
- To enable users to perform **early risk assessment for common diseases**.
- To provide a **simple and user-friendly health monitoring platform**.

3. Problem Statement

Many existing hospital software systems are primarily designed for hospital staff and administrative management. These systems rarely provide intelligent health monitoring features that allow patients to actively check their health status.

Furthermore, early detection of diseases is often difficult because individuals do not have access to quick screening tools without visiting a hospital. This results in delayed diagnosis and increased health risks.

Another issue is that many individuals do not regularly track important health indicators such as BMI, weight, or disease risk factors.

Therefore, there is a need for a **smart mobile application that allows users to monitor their health status and predict potential disease risks using artificial intelligence techniques**. This project aims to address this gap by developing a multi-disease prediction system integrated with a health monitoring dashboard.

4. Related Work

Several research studies and healthcare applications have attempted to integrate machine learning with healthcare systems.

For example, machine learning techniques have been widely used to predict diseases such as heart disease, diabetes, and liver disorders using datasets available in public repositories such as the **UCI Machine Learning Repository** and **Kaggle**.

Previous studies have applied algorithms such as Logistic Regression, Random Forest, Support Vector Machine (SVM), and Neural Networks for disease prediction tasks.

However, many of these systems remain limited to web-based research prototypes and do not provide a fully integrated **mobile healthcare monitoring solution with multiple disease prediction features**.

This project aims to combine **machine learning prediction models with a mobile healthcare dashboard** that allows users to easily interact with the system and monitor their health data.

5. Scope

The scope of the project includes the following functionalities:

- Development of a **Flutter-based mobile application**.
- Creation of a **patient health profile system**.

- Implementation of a **BMI calculator** based on height and weight.
- Integration of **machine learning models for disease prediction**.
- Support for predicting multiple diseases such as:
 - Heart Disease
 - Diabetes
 - Liver Disease
 - Kidney Disease
 - Pneumonia
- Visualization of prediction results and health status.
- Secure storage and management of patient health data.

The application will focus on **preliminary risk assessment rather than clinical diagnosis**.

6. Methodology

The project will follow a structured development methodology consisting of data collection, model development, application development, and system integration.

The methodology will include both **machine learning model development** and **mobile application implementation**.

6.1 Step-by-Step Development Methodology

The development process will include the following steps:

Step 1: Requirement Analysis

Identify system requirements, user needs, and healthcare features.

Step 2: Dataset Collection

Collect publicly available medical datasets from platforms such as Kaggle and UCI Machine Learning Repository.

Step 3: Data Preprocessing

Clean and preprocess datasets by handling missing values, normalization, and feature selection.

Step 4: Machine Learning Model Development

Train machine learning models such as Random Forest, Logistic Regression, and Neural Networks for disease prediction.

Step 5: Model Evaluation

Evaluate model performance using metrics such as accuracy, precision, recall, and confusion matrix.

Step 6: Mobile Application Development

Develop the mobile application using Flutter with user-friendly UI components.

Step 7: API Integration

Integrate the trained machine learning models with the mobile application through APIs.

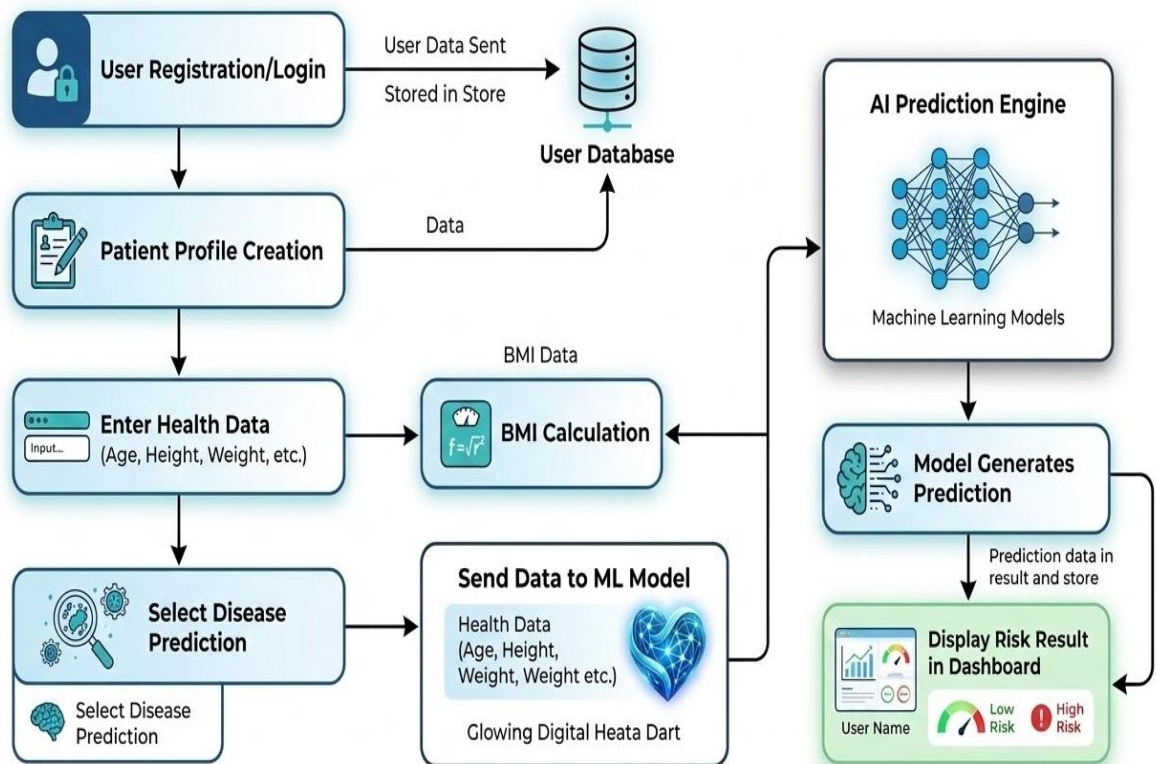
Step 8: Testing and Validation

Perform system testing to ensure correct predictions and smooth user interaction.

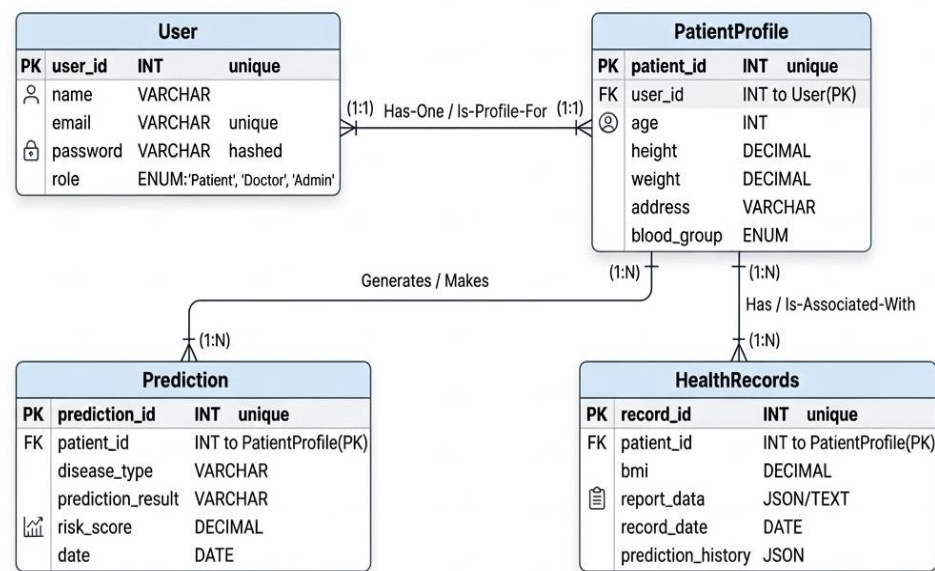
7. Visual Models

Visual models help represent the structure and functionality of the system.

7.1 Flow Chart Diagram



7.2 ERD (Entity Relationship Diagram)



7.3 Timeline (Gantt Chart)

Task	Week 1-2	Week 3-4	Week 5-6	Week 7-8	Week 9	Week 10	Week 11	Week 12
Requirements Analysis & UI Mockup	✓		✓					
App Structure & Navigation Design		✓	✓					
Patient Profile & BMI Feature Development			✓	✓				
Disease Prediction Model Integration			✓	✓	✓			
Database & Backend Integration				✓	✓			
Health Dashboard & Result Visualization					✓	✓		
UI Improvement & Documentation						✓	✓	
Final Testing & Deployment								✓

8. Limitations

The proposed system has some limitations:

- The predictions are **not a substitute for professional medical diagnosis**.
- Accuracy depends on the **quality of the dataset and model training**.
- The system may not cover **all possible diseases**.
- Real clinical data integration may be limited due to privacy issues.

9. Future Plans

Future improvements may include:

- Integration with **hospital databases and electronic health records**.
- Adding **more disease prediction models**.
- Integration of **wearable health device data**.
- Implementation of **real-time health monitoring**.
- Adding **AI chatbot for health assistance**.

10. Result

The expected result of this project is a functional mobile application that allows users to:

- Maintain their health profile
- Calculate BMI automatically
- Predict the risk of multiple diseases
- Monitor their health status through a simple dashboard

The system will demonstrate the practical application of **machine learning in healthcare monitoring** and provide a user-friendly mobile solution for early disease risk detection.

11. References

1. UCI Machine Learning Repository – Medical datasets
2. Kaggle – Disease prediction datasets
3. Flutter Official Documentation
4. Research papers on Machine Learning based Disease Prediction